



Psoriasis

By [**Shinjita Das**](#), MD MPH, Massachusetts General Hospital
Reviewed/Revised Nov 2025

Psoriasis is a chronic, recurring disease that causes one or more raised, red patches that have silvery scales and a distinct border between the patch and normal skin.

- A problem with the immune system may play a role, and some people are genetically predisposed to psoriasis.
- Characteristic scales or red patches can appear anywhere on the body in large or small patches, particularly the elbows, knees, and scalp.
- This disease is treated with a combination of medications applied to the skin, exposure to ultraviolet light (phototherapy), and medications taken by mouth or given by injection.

Psoriasis is common and varies widely among different populations worldwide. It is likely that psoriasis is underreported among people with dark skin compared with people with light skin.

Psoriasis is found more often in adults than children, but can occur at any age

The patches of psoriasis occur because of an abnormally high rate of growth of skin cells. The reason for the rapid cell growth is unknown, but a problem with the immune system is thought to play a role. The disorder often runs in families, and certain genes are associated with psoriasis.

Some well-identified triggers of psoriasis include:

- Minor skin injuries
- Sunburn
- HIV infection
- Streptococcal infection, such as strep throat (leading to guttate psoriasis)
- Medications (especially beta-blockers, [chloroquine](#), [lithium](#), angiotensin-converting enzyme inhibitors, [indomethacin](#), [terbinafine](#), interferon-alfa, immune checkpoint inhibitors, and tumor necrosis factor inhibitors)
- Emotional stress
- Alcohol consumption
- Tobacco smoking

- Obesity

Symptoms of Psoriasis

Red Patch in Psoriasis

IMAGE



Psoriasis typically causes red bumps or patches covered with silvery, shiny scales.

IMAGE PROVIDED BY THOMAS HABIF, MD.

Plaque Psoriasis

IMAGE



This photo shows plaques of psoriasis on dark skin. Plaques may look bluish purple in people with dark skin tones, but key features such as scaliness and location on the body (for example, on the elbows, trunk, legs, and around the navel) are consistent across all skin tones.

IMAGE COURTESY OF KAREN MCKOY, MD.

Plaque psoriasis, the most common type of psoriasis, usually starts as one or more small red, silvery, shiny patches (plaques) on the scalp, elbows, knees, back, or buttocks. The eyebrows, underarms, navel, the skin around the anus, and the cleft where the buttocks meet the lower back may also be affected. Many people with psoriasis may also have deformed, thickened, and pitted nails.

Nail Psoriasis With Pits and Discoloration

Hide Details

This photo shows pits and areas of tan-brown discoloration (called oil spots) in a person who has psoriasis.

© Springer Science+Business Media

The first patches may clear up after a few months or remain, sometimes growing together to form larger patches. Some people never have more than one or two small patches, and others have patches covering large areas of the body. Thick patches or patches on the palms of the hands, soles of the feet, or skinfolds of the genitals are more likely to itch or hurt, but many times the person has no symptoms. Although the patches do not cause extreme physical discomfort, they are very obvious and often embarrassing to the person. The psychological distress caused by psoriasis can be severe.

Psoriasis persists throughout life but may come and go. Symptoms of psoriasis are often diminished during the summer when the skin is exposed to bright sunlight. Some people may go for years between occurrences.

Psoriatic Arthritis of the Fingers

Hide Details

This photo shows red, scaly plaques that are characteristic of psoriatic arthritis. The nails are yellowed and the texture has changed.

DR. HAROUT TANIELIAN/SCIENCE PHOTO LIBRARY

About 5 to 30% of people with psoriasis develop arthritis ([psoriatic arthritis](#)). Psoriatic arthritis causes joint pain and swelling.

Flare-ups

Psoriasis may flare up for no apparent reason or as a result of a variety of circumstances. Flare-ups often result from conditions that irritate the skin, such as minor injuries and severe [sunburn](#). Sometimes flare-ups occur after infections, such as colds and [strep throat](#). Flare-ups are more common in the winter, after drinking alcohol, and after stressful situations. Many medications, such as antimalarial medications, [lithium](#), angiotensin-converting enzyme (ACE) inhibitors, [terbinafine](#), interferon-alpha, beta-blockers, and tumor necrosis factor inhibitors, can also cause psoriasis to flare up. Flare-ups are also more common among people with [obesity](#), infected with the [human immunodeficiency virus](#) (HIV), or smoke tobacco.

Uncommon types of psoriasis

Some uncommon types of psoriasis can have more serious effects.

Erythrodermic psoriasis causes all of the skin on the body to become red and scaly. This form of psoriasis is serious because, like a burn, it keeps the skin from serving as a protective barrier against injury and infection.

Erythrodermic Psoriasis

Hide Details

This photo shows a widespread red rash on the torso of a person who has erythrodermic psoriasis.

SCIENCE PHOTO LIBRARY

Pustular psoriasis is another uncommon form of psoriasis. In this form, large and small pus-filled blisters (pustules) are scattered widely on the body.

Pustular Psoriasis (Feet)

Hide Details

This image shows widespread pus-filled blisters (pustules) on the soles of the feet of a person with pustular psoriasis.

Image courtesy of Karen McKoy, MD.

Palmoplantar psoriasis is a form of pustular psoriasis in which pustules occur primarily on the hands and feet. It is sometimes called palmoplantar psoriasis of the palms and soles.

Palmoplantar Psoriasis of the Palm

Hide Details

This photo shows collections of rough, scaly skin on the palm of a person who has palmoplantar psoriasis. The collections are what remain after pus-filled blisters (pustules) have ruptured on their own.

SCIENCE PHOTO LIBRARY

Palmoplantar Psoriasis of the Sole

Hide Details

This photo shows rough, scaly skin on the sole of a person who has palmoplantar psoriasis. The scales are what remain on the skin after pus-filled blisters (pustules) have ruptured on their own.

SCIENCE PHOTO LIBRARY

Guttate psoriasis is an uncommon form of psoriasis in which multiple, small patches (plaques) appear suddenly, usually on the trunk in children and young adults after a [strep throat](#) infection. Antibiotics to treat the strep infection make the patches go away in some people, but others still need additional therapies.

Guttate Psoriasis

Hide Details

This photo shows multiple patches (plaques) on the trunk of a 22-year-old man with guttate psoriasis.

DR P. MARAZZI/SCIENCE PHOTO LIBRARY

Diagnosis of Psoriasis

- A doctor's evaluation
- Sometimes skin biopsy

Doctors base the diagnosis of psoriasis on how the scales and plaques look and where they appear on the body.

Sometimes, doctors take a sample of skin tissue and examine it under a microscope to rule out other disorders (such as [skin cancer](#)).

Treatment of Psoriasis

- Topical treatments
- Phototherapy
- Systemic treatments

Many medications are available to treat psoriasis. Most often, a combination of medications is used, depending on the severity and extent of the person's symptoms.

Topical treatments

Topical treatments are applied to the skin. They are the most common treatments for psoriasis. Nearly everyone with psoriasis benefits from [skin moisturizers](#) (emollients).

Other topical agents include [steroids](#) (sometimes also known as glucocorticoids or corticosteroids), often used together with [calcipotriene](#) (also called calcipotriol, which is a form of [vitamin D](#)), or [coal tar](#).

[Tacrolimus](#) and [pimecrolimus](#) are used to treat psoriasis that appears on delicate skin (such as on the face or groin or in skinfolds). [Tazarotene](#), [roflumilast](#), or [tapinarof](#) may also be used.

Very thick patches can be thinned with ointments containing [salicylic acid](#), which make the other treatments more effective.

[Coal tar](#) and [anthralin](#) are other treatment options, but they are being used less frequently.

Many of these treatments are irritating to the skin, and doctors must find which ones work best for each person.

Phototherapy

Phototherapy (exposure to ultraviolet light) also can help clear up psoriasis for several months at a time (see sidebar [Phototherapy: Using Ultraviolet Light to Treat Skin Disorders](#)). Phototherapy is often used in combination with various topical medications, particularly when large areas of skin are involved, but it is not being used as much because other effective medications are available. Traditionally, treatment has been with phototherapy combined with the use of psoralens (medications that make the skin more sensitive to the effects of ultraviolet light). This treatment is called PUVA (psoralen plus ultraviolet A).

Many doctors are now using narrowband ultraviolet B (NBUVB) treatments, which are as effective as PUVA. However, NBUVB treatments are done without psoralens and therefore do not have the same side effects, such as extreme sensitivity to sunlight.

Doctors can also treat specific patches of the skin directly by using a laser that focuses ultraviolet light (called excimer laser therapy).

Phototherapy: Using Ultraviolet Light to Treat Skin Disorders

Exposure to sunlight is helpful for certain skin disorders. One component of sunlight—ultraviolet (UV) light—is responsible for this effect. UV light has many different effects on skin cells, including altering the amounts and kinds of chemicals they make and causing the death of certain cells that can be involved in skin diseases.

The use of UV light to treat disease is called phototherapy. [Psoriasis](#) and [atopic dermatitis](#) are the disorders most commonly treated with phototherapy.

Because exposure to natural sunlight varies in intensity and is not practical for a large part of the year in certain climates, phototherapy is nearly always done with artificial UV light. Treatments are given in a doctor's office or in a specialized treatment center.

UV light, which is invisible to the human eye, is classified as A, B, or C, depending on its wavelength. Ultraviolet A (UVA) penetrates deeper into the skin than ultraviolet B (UVB). UVA or UVB is chosen based on the type and severity of the person's disorder. Ultraviolet C is not used in phototherapy. Some lights produce only certain specific wavelengths of UVA or UVB (narrowband therapy), which are used to treat specific disorders. Narrowband therapy helps limit the sunburn-like effect caused by phototherapy.

Phototherapy is sometimes combined with the use of a psoralen. The combination of a psoralen plus UVA is known as PUVA therapy. Psoralens are medications that may be taken by mouth before treatment with UV light. Psoralens sensitize the skin to the effects of UV light, allowing shorter, less intense exposure.

Side effects of phototherapy include pain and reddening similar to sunburn with prolonged exposure to UV light. UV light exposure also increases the long-term risk of skin cancer, but the risk is small for brief courses of treatment. Psoralens often cause nausea and extreme sensitivity to sunlight. In addition, because psoralens enter the lens of the eye, UV-resistant sunglasses must be worn for at least 12 hours after undergoing PUVA therapy.

Systemic treatments

Systemic treatments are medications that affect the whole body or certain systems of the body. They can be taken by mouth or injected.

Systemic immunosuppressants are medications that intentionally weaken (suppress) the immune system to keep it from making psoriasis worse. Immunosuppressants can reduce the body's ability to fight infections. Examples include:

- **Cyclosporine** can be used to treat severe psoriasis. This medication may cause high blood pressure and damage the kidneys.
- **Methotrexate** decreases inflammation in the body and interferes with the growth and multiplication of skin cells. Doctors use [methotrexate](#) to treat people whose psoriasis is severe or does not respond to less harmful forms of therapy. Liver damage and impaired immunity are possible side effects.
- **Biologic agents** are made from living organisms and inhibit certain chemicals involved in the immune system. They include [etanercept](#), [adalimumab](#), [infliximab](#), [certolizumab pegol](#), [ustekinumab](#), [secukinumab](#), [brodalumab](#), [ixekizumab](#), [tildrakizumab](#), [risankizumab](#), [guselkumab](#), and [tofacitinib](#). All of these medications are given by injection except [tofacitinib](#), which is taken by mouth. These medications tend to be the most effective ones for severe psoriasis, but they can have more troublesome side effects. Biosimilar medications are medications that are highly similar to biologic agents and are sometimes substituted for biologic agents.

Other medications may be given to treat moderate to severe forms of psoriasis and psoriatic arthritis:

- **Systemic retinoids**, such as [acitretin](#) and [isotretinoin](#), may be effective for severe and hard to treat cases of plaque psoriasis, pustular psoriasis (for which isotretinoin is often preferred), and palmoplantar psoriasis. These medications are given by mouth. They can cause severe birth defects and should not be taken by women who may become pregnant. Women should wait at least 3 years after their last dose of acitretin and at least 1 month after their last dose of isotretinoin before attempting pregnancy. These medications can raise levels of triglycerides and cause reversible problems with the liver, blood counts, bones, and hair.
- **Apremilast** is another option and it is taken by mouth. The most common side effects are nausea and diarrhea.

More Information

The following English-language resource may be useful. Please note that The Manual is not responsible for the content of this resource.

[National Psoriasis Foundation](#): A resource providing information about treatments and community support

Drug Information for the Topic
